M3503 WEEK 8 2022

LINEAR SYSTEMS OF DIFFERENTIAL EQUATIONS

The system

$$\begin{aligned} y_1' &= a_{11}(t)y_1 + a_{12}(t)y_2 + \cdots + a_{1n}(t)y_n + f_1(t) \\ y_2' &= a_{21}(t)y_1 + a_{22}(t)y_2 + \cdots + a_{2n}(t)y_n + f_2(t) \\ &\vdots \\ y_n' &= a_{n1}(t)y_1 + a_{2n}(t)y_2 + \cdots + a_{nn}(t)y_n + f_n(t) \end{aligned}$$

is an $n \times n$ first order linear system. When $f_i(t) = 0$, $i = 1, 2, \ldots n$, the linear system is homogeneous, otherwise, it is nonhomogeneous.

High order differential equations can be written as systems of first order equations.

Example

Write $y^{(4)} + 4y''' + 6y'' + 4y' + y = 0$ as a first order system.

Soln

We take y''', y'', y' and y as unknown functions, and let

$$y = y_1 \quad , \quad y' = y_2, \quad y'' = y_3 \quad , \quad y''' = y_4 ,$$

so that the given equation can be written as

$$y_4' + 4y_4 + 6y_3 + 4y_2 + y_1 = 0$$

The functions y_1, y_2, y_3 and y_4 satisfy the system

$$\begin{aligned} y_1' &= y_2 \\ y_2' &= y_3 \\ y_3' &= y_4 \\ y_4' &= -4y_4 - 6y_3 - 4y_2 - y_1 \end{aligned}$$

MATRIX FORM OF LINEAR SYSTEMS

Let $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$, $A(t) = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & & \vdots \\ a_{n1}(t) & a_{n2}(t) & & a_{nn}(t) \end{bmatrix}$, $f(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{bmatrix}$

The matrix form of the linear system is

$$\begin{bmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{bmatrix} = \begin{bmatrix} a_{11}(t) & a_{12}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \cdots & a_{2n}(t) \\ \vdots & \vdots & & \vdots \\ a_{n1}(t) & a_{n2}(t) & \cdots & a_{nn}(t) \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} + \begin{bmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{bmatrix}$$

or $y' = A(t)\,y + f(t)$

Examples

① Write the system in matrix form

(a) $y_1' = 3y_1 - 5y_2$

$y_2' = 4y_1 + 8y_2$

(b) $y_1' = y_1 - y_2 + y_3 + t - 1$

$y_2' = 2y_1 + y_2 - y_3 - 3t^2$

$y_3' = y_1 + y_2 + y_3 + t^2 - t + 2$

② Verify that $y = \begin{bmatrix} 5\cos t \\ 3\cos t - \sin t \end{bmatrix} e^t$ is a solution to the system $y_1' = -2y_1 + 5y_2$

$y_2' = -2y_1 + 4y_2$

Sol^n

① (a) $y' = \begin{bmatrix} 3 & -5 \\ 4 & 8 \end{bmatrix} y$

(b) $y' = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} y + \begin{bmatrix} t-1 \\ -3t^2 \\ t^2-t+2 \end{bmatrix}$

② . $y' = \begin{bmatrix} -2 & 5 \\ -2 & 4 \end{bmatrix} y$

$$y = \begin{bmatrix} 5\cos t \\ 3\cos t - \sin t \end{bmatrix} e^t \Rightarrow y' = \begin{bmatrix} -5\sin t + 5\cos t \\ 2\cos t - 4\sin t \end{bmatrix} e^t$$

Substituting y into the system gives

$$\begin{bmatrix} -2 & 5 \\ -2 & 4 \end{bmatrix} y = \begin{bmatrix} -2 & 5 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 5\cos t \\ 3\cos t - \sin t \end{bmatrix} e^t$$

$$= \begin{bmatrix} -10\cos t + 15\cos t - 5\sin t \\ -10\cos t + 12\cos t - 4\sin t \end{bmatrix} e^t$$

$$= \begin{bmatrix} 5\cos t - 5\sin t \\ 2\cos t - 4\sin t \end{bmatrix}$$

$$= y'$$

<u>Theorem</u> : Existence of a unique solution
If matrix A and function f are continuous on (a,b), and t_0 lies in (a,b), then the initial value problem

$$y' = A(t)\,y + f(t), \quad y(t_0) = y_0$$

has a unique solution on (a,b)

The theory of linear systems is much similar to the theory of nth-order differential equations.

Theorem: Superposition principle for homogeneous systems

If $y_1, y_2, \ldots, y_n$ are solutions to $y' = A(t)y$ on (a,b), then $y = c_1y_1 + c_2y_2 + \cdots + c_ny_n$ is a solution to $y' = A(t)y$.

LINEAR DEPENDENCE/INDEPENDENCE

Let $y_1, y_2, \ldots, y_n$ be solutions to $y' = A(t)y$ on (a,b). The set $\{y_1, y_2, \ldots, y_n\}$ is linearly dependent on (a,b) if

$$c_1y_1 + c_2y_2 + \cdots + c_ny_n = 0$$

with constants $c_1, c_2, \ldots, c_n$ not all zero, otherwise it is linearly independent.

CRITERION FOR LINEARLY INDEPENDENT SOLUTIONS

Let $y_1 = \begin{bmatrix} y_{11} \\ y_{21} \\ \vdots \\ y_{n1} \end{bmatrix}$, $y_2 = \begin{bmatrix} y_{12} \\ y_{22} \\ \vdots \\ y_{n2} \end{bmatrix}$, $\ldots$, $y_n = \begin{bmatrix} y_{1n} \\ y_{2n} \\ \vdots \\ y_{nn} \end{bmatrix}$

be solution vectors of $y' = A(t)y$ on (a,b). The set $\{y_1, y_2, \ldots, y_n\}$ is linearly independent if the Wronskian is non-zero on (a,b).

i.e $W(y_1, y_2, \ldots, y_n) = \begin{vmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nn} \end{vmatrix} \neq 0$ on (a,b).

Examples

Determine if the given vectors are linearly independent on $(-\infty, \infty)$

① $y_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-2t}$, $y_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-6t}$

② $y_1 = \begin{bmatrix} \cos t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t \\ -\cos t - \sin t \end{bmatrix}$, $y_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{t}$, $y_3 = \begin{bmatrix} \sin t \\ -\frac{1}{2}\sin t - \frac{1}{2}\cos t \\ -\sin t + \cos t \end{bmatrix}$

Solution

① $W(y_1, y_2) = \begin{vmatrix} e^{-2t} & e^{-6t} \\ e^{-2t} & -e^{-6t} \end{vmatrix}$

$= -e^{-8t} - e^{-8t}$

$= -2e^{-8t} \neq 0$ on $(-\infty, \infty)$

$\therefore y_1$ and y_2 are linearly independent on $(-\infty, \infty)$.

② $W(y_1, y_2, \ldots, y_n) = \begin{vmatrix} \cos t & 0 & \sin t \\ -\frac{1}{2}\cos t + \frac{1}{2}\sin t & e^{t} & -\frac{1}{2}\sin t - \frac{1}{2}\cos t \\ -\cos t - \sin t & 0 & -\sin t + \cos t \end{vmatrix}$

$= e^{t} \begin{vmatrix} \cos t & \sin t \\ -\cos t - \sin t & -\sin t + \cos t \end{vmatrix}$

$= e^{t}[-\sin t \cos t + \cos^2 t + \sin t \cos t + \sin^2 t]$

$= e^{t} \neq 0$ on $(-\infty, \infty)$

$\therefore y_1, y_2$ and y_3 are linearly independent on $(-\infty, \infty)$

Homogeneous Linear Systems

To solve the homogeneous system

$$y' = A(t)\,y,$$

we determine the eigenvalues of A, and their corresponding eigenvectors. Eigenvalues may be distinct, repeated or complex.

Distinct Real Eigenvalues

If nxn coefficient matrix A has n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, with corresponding eigenvectors $V_1, V_2, \ldots, V_n$, then the general solution of $y' = A(t)\,y$ on $(-\infty, \infty)$ is

$$y = c_1 V_1 e^{\lambda_1 t} + c_2 V_2 e^{\lambda_2 t} + \cdots + c_n V_n e^{\lambda_n t}.$$

Examples

Solve

① $y_1' = y_1 + 2y_2$

$y_2' = +y_1 + 3y_2$

② $y_1' = -4y_1 + y_2 + y_3$

$y_2' = y_1 + 5y_2 - y_3$

$y_3' = y_2 - 3y_3$

Soln

① $y' = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} y = Ay$

We first find the eigenvalues of A using the characteristic equation $\det(A - \lambda I) = 0$, where I is the identity matrix, and λ is the eigenvalue of A.

$$\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = 0$$

$\Rightarrow (1-\lambda)(3-\lambda) - 8 = 0$

$\Rightarrow \lambda^2 - 4\lambda - 5 = 0$

$\Rightarrow (\lambda - 5)(\lambda + 1) = 0$

$\Rightarrow \lambda = 5, \quad \lambda = -1$

Next we find the eigenvector corresponding to each eigenvalue by solving for V in

$$(A - \lambda I)V = 0$$

For $\lambda = 5$, we have

$$\begin{pmatrix} -4 & 2 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using row operations, we get

$$\left(\begin{array}{cc|c} -4 & 2 & 0 \\ 4 & -2 & 0 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow V_1 - \frac{1}{2}V_2 = 0 \Rightarrow 2V_1 = V_2$

So $\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} p \\ 2p \end{pmatrix}$, where $p \in \mathbb{R}$

Taking $P = 1$, we get the eigenvector as

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

For $\lambda=-1$, we have

$$\left(\begin{array}{cc|c} 2 & 2 & 0 \\ 4 & 4 & 0 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow u_1+u_2=0 \Rightarrow u_2=-u_1$

So, $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} p \\ -p \end{pmatrix}$, where $p \in \mathbb{R}$

$\therefore \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is the eigenvector corresponding to $\lambda=-1$

$$\therefore y = c_1\begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{5t} + c_2\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t}$$

(2). $$y' = \begin{pmatrix} -4 & 1 & 1 \\ 1 & 5 & -1 \\ 0 & 1 & -3 \end{pmatrix} y$$

$$\begin{vmatrix} -4-\lambda & 1 & 1 \\ 1 & 5-\lambda & -1 \\ 0 & 1 & -3-\lambda \end{vmatrix} = 0$$

$\Rightarrow -(4+\lambda)[(5-\lambda)(-3-\lambda)+1] - [-3-\lambda-1] = 0$

$\Rightarrow (4+\lambda)[(5-\lambda)(3+\lambda)-1] + (4+\lambda) = 0$

$\Rightarrow (4+\lambda)(5-\lambda)(3+\lambda) = 0$

$\Rightarrow \lambda=-4, \lambda=5, \lambda=-3$

For $\lambda=-4$, we have

$$\left(\begin{array}{ccc|c} 0 & 1 & 1 & 0 \\ 1 & 9 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 9 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & -10 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow v_1 - 10v_3 = 0 \Rightarrow v_1 = 10v_3$

and $v_2+v_3=0 \Rightarrow v_2=-v_3$

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 10p \\ -p \\ p \end{pmatrix}, \quad p \in \mathbb{R}$$

$\therefore \begin{pmatrix} 10 \\ -1 \\ 1 \end{pmatrix}$ is the eigenvector for $\lambda = -4$

For $\lambda = 5$, we have

$$\left(\begin{array}{ccc|c} -9 & 1 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -8 & 0 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -8 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow u_1 - u_3 = 0 \Rightarrow u_1 = u_3$

and $u_2 - 8u_3 = 0 \Rightarrow u_2 = 8u_3$

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} p \\ 8p \\ p \end{pmatrix}, \quad p \in \mathbb{R}$$

$\therefore \begin{pmatrix} 1 \\ 8 \\ 1 \end{pmatrix}$ is the eigenvector for $\lambda = 5$

For $\lambda = -3$, we have

$$\left(\begin{array}{ccc|c} -1 & 1 & 1 & 0 \\ 1 & 8 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow w_2 = 0$

and $w_1 - w_2 - w_3 = 0 \Rightarrow w_1 = w_3$

$$\begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} p \\ 0 \\ p \end{pmatrix}, \quad p \in \mathbb{R}$$

$\therefore \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is the eigenvector for $\lambda = -3$

$$\therefore y = C_1 \begin{pmatrix} 10 \\ -1 \\ 1 \end{pmatrix} e^{-4t} + C_2 \begin{pmatrix} 1 \\ 8 \\ 1 \end{pmatrix} e^{5t} + C_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{-3t}$$

REPEATED EIGENVALUES

If λ is an eigenvalue of multiplicity m, we can have at most m linearly independent eigenvectors corresponding to λ. For $n \times n$ matrix A

① if m linearly independent eigenvectors $K_1, K_2, \ldots, K_m$ corresponding to λ of multiplicity $m \leq n$ can be found, then the general solution of the system contains the linear combination

$$C_1 K_1 e^{\lambda t} + C_2 K_2 e^{\lambda t} + \cdots + C_m K_m e^{\lambda t}$$

② if only one eigenvector corresponding to λ is found, then m linearly independent solutions of the form

$$y_1 = K_{11} e^{\lambda t}$$

$$y_2 = K_{21} t e^{\lambda t} + K_{22} e^{\lambda t}$$

$$\vdots$$

$$y_m = K_{m1} \frac{t^{m-1}}{(m-1)!} e^{\lambda t} + K_{m2} \frac{t^{m-2}}{(m-2)!} e^{\lambda t} + \cdots + K_{mm} e^{\lambda t},$$

where K_{ij} are column vectors, can be found.

Examples

solve

① $y' = \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix} y$

② $y' = \begin{pmatrix} 5 & 1 \\ -4 & 1 \end{pmatrix} y$

③ $y' = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & -1 \\ 0 & 1 & 0 \end{pmatrix} y$

Sol^{n}

① $\begin{vmatrix} 3-\lambda & 2 & 4 \\ 2 & -\lambda & 2 \\ 4 & 2 & 3-\lambda \end{vmatrix} = 0$

$\Rightarrow (3-\lambda)[\lambda(\lambda-3)-4] - 2[2(3-\lambda)-8] + 4(4+4\lambda) = 0$

$\Rightarrow (3-\lambda)(\lambda-4)(\lambda+1) + 4(\lambda+1) + 16(\lambda+1) = 0$

$\Rightarrow (\lambda+1)[(3-\lambda)(\lambda-4)+20] = 0$

$\Rightarrow (\lambda+1)[\lambda^2 - 7\lambda - 8] = 0$

$\Rightarrow (\lambda+1)(\lambda-8)(\lambda+1) = 0$

$\Rightarrow (\lambda+1)^2(\lambda-8) = 0$

$\Rightarrow \lambda = -1, \quad \lambda = 8$

For $\lambda = 8$, we have

$$\left(\begin{array}{ccc|c} -5 & 2 & 4 & 0 \\ 2 & -8 & 2 & 0 \\ 4 & 2 & -5 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & -2/5 & -4/5 & 0 \\ 0 & -36 & 18 & 0 \\ 0 & -18 & 9 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & -2/5 & -4/5 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow v_2 - \frac{1}{2}v_3 = 0 \quad \Rightarrow v_2 = \frac{1}{2}v_3$

and $v_1 - \frac{2}{5}v_2 - \frac{4}{5}v_3 = 0 \quad \Rightarrow v_1 = v_3$

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} p \\ \frac{1}{2}p \\ p \end{pmatrix}$$

$\therefore \begin{pmatrix} 1 \\ \frac{1}{2} \\ 1 \end{pmatrix}$ is the eigenvector for $\lambda = 8$

For $\lambda = -1$, we have

$$\left(\begin{array}{ccc|c} 4 & 2 & 4 & 0 \\ 2 & 1 & 2 & 0 \\ 4 & 2 & 4 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow u_1 + \frac{1}{2}u_2 + u_3 = 0$

$\Rightarrow \quad u_1 = -u_3 - \frac{1}{2}u_2$

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}p \\ p \\ 0 \end{pmatrix} + \begin{pmatrix} -s \\ 0 \\ s \end{pmatrix} \quad , s \in \mathbb{R}, p \in \mathbb{R}$$

$$\therefore \begin{pmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \text{ are eigenvectors of } \lambda = -1$$

$$\therefore y = c_1 \begin{pmatrix} 1 \\ \frac{1}{2} \\ 1 \end{pmatrix} e^{8t} + c_2 \begin{pmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{-t}$$

② $$\begin{vmatrix} 5-\lambda & 1 \\ -4 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (5-\lambda)(1-\lambda) + 4 = 0$$
$$\Rightarrow \lambda^2 - 6\lambda + 5 + 4 = 0$$
$$\Rightarrow \lambda^2 - 6\lambda + 9 = 0$$
$$\Rightarrow (\lambda - 3)^2 = 0$$
$$\Rightarrow \lambda = 3$$

$$\left(\begin{array}{cc|c} 2 & 1 & 0 \\ -4 & -2 & 0 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$$\Rightarrow V_1 = -\tfrac{1}{2} V_2$$

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}p \\ p \end{pmatrix} \quad ; p \in \mathbb{R}$$

$$\therefore \begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix} \text{ is the eigenvector corresponding to } \lambda = 3$$

Since $\lambda = 3$ is of multiplicity 2, and we only have one eigenvector, we look for the second vector

using the equation

$$(A-\lambda I)u=v$$

So, we have

$$\left(\begin{array}{cc|c} 2 & 1 & -\frac{1}{2} \\ -4 & -2 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & \frac{1}{2} & -\frac{1}{4} \\ 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow u_1 + \frac{1}{2}u_2 = -\frac{1}{4}$

$\Rightarrow u_1 = -\frac{1}{4} - \frac{1}{2}u_2$

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}p - \frac{1}{4} \\ p \end{pmatrix}, \quad p \in \mathbb{R}$$

Taking $p=0$, the second vector is

$$\begin{pmatrix} -\frac{1}{4} \\ 0 \end{pmatrix}$$

$$\therefore y = C_1\begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix}e^{3t} + C_2\left[\begin{pmatrix} -\frac{1}{2} \\ 1 \end{pmatrix}te^{3t} + \begin{pmatrix} -\frac{1}{4} \\ 0 \end{pmatrix}e^{3t}\right]$$

③.
$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 2 & 2-\lambda & -1 \\ 0 & 1 & -\lambda \end{vmatrix} = 0$$

$\Rightarrow (1-\lambda)[(2-\lambda)\lambda - 1] = 0$

$\Rightarrow (1-\lambda)[-\lambda^2 + 2\lambda - 1] = 0$

$\Rightarrow (1-\lambda)^3 = 0$

$\Rightarrow \lambda = 1$

$$\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 2 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow V_2 = V_3$

and $V_1 + \frac{1}{2}V_2 - \frac{1}{2}V_3 = 0 \Rightarrow V_1 = 0$

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ p \\ p \end{pmatrix}, \quad p \in \mathbb{R}$$

$\therefore \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ is the eigenvector for $\lambda = 1$

To find the other vector, we solve

$$(A - \lambda I)u = v \quad , \text{ so}$$

$$\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 2 & 1 & -1 & 1 \\ 0 & 1 & -1 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\Rightarrow u_1 = 0$$

and $u_2 = u_3 + 1$

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 0 \\ p+1 \\ p \end{pmatrix}, \quad p \in \mathbb{R}$$

Taking $p = 0$, the ~~vector~~ vector is $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

To find the third vector, we solve

$$(A - \lambda I)w = u \quad , \text{ so}$$

$$\left(\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 2 & 1 & -1 & 1 \\ 0 & 1 & -1 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\Rightarrow w_1 = \frac{1}{2} \quad \text{and} \quad w_2 = w_3$$

$$\begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ p \\ p \end{pmatrix}, \quad p \in \mathbb{R}$$

Taking $p = 0$, the third vector is $\begin{pmatrix} \frac{1}{2} \\ 0 \\ 0 \end{pmatrix}$

$$\therefore y = c_1\begin{pmatrix}0\\1\\1\end{pmatrix}e^t + c_2\left[\begin{pmatrix}0\\1\\1\end{pmatrix}te^t + \begin{pmatrix}0\\1\\0\end{pmatrix}e^t\right] + c_3\left[\begin{pmatrix}0\\1\\1\end{pmatrix}t^2e^t + \begin{pmatrix}0\\1\\0\end{pmatrix}te^t + \begin{pmatrix}1/2\\0\\0\end{pmatrix}e^t\right]$$

Complex Eigenvalues

If the coefficient matrix has complex eigenvalue $\lambda = \alpha + i\beta$, with corresponding eigenvector K, the solution of $y' = Ay$ contains the linear combination $c_1 y_1 + c_2 y_2$, where

$$y_1 = [Re(K)\cos(\beta t) - Im(K)\sin(\beta t)]\,e^{\alpha t}$$

$$y_2 = [Im(K)\cos(\beta t) + Re(K)\sin(\beta t)]\,e^{\alpha t}$$

Examples

Solve

① $y' = \begin{pmatrix}6 & -1\\ 5 & 2\end{pmatrix} y$

② $y' = \begin{pmatrix}0 & 0 & 1\\ 0 & 0 & -1\\ 0 & 1 & 0\end{pmatrix} y$

Sol^{n}

① $\begin{vmatrix}6-\lambda & -1\\ 5 & 2-\lambda\end{vmatrix} = 0$

$\Rightarrow (6-\lambda)(2-\lambda) + 5 = 0$

$\Rightarrow \lambda^2 - 8\lambda + 12 + 5 = 0$

$\Rightarrow \lambda^2 - 8\lambda + 17 = 0$

$$\Rightarrow \lambda = \frac{8 \pm \sqrt{64 - 4(17)}}{2}$$

$$= 4 \pm i$$

Taking $\lambda = 4+i$, we have

$$\left(\begin{array}{cc|c} 2-i & -1 & 0 \\ 5 & -2-i & 0 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & -\frac{1}{2-i} & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$$\Rightarrow u_1 - \frac{1}{2-i} u_2 = 0$$

$$\Rightarrow (2-i)\, u_1 = u_2$$

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} p \\ (2-i)p \end{pmatrix}, \quad p \in \mathbb{R}$$

$$\therefore \begin{pmatrix} 1 \\ 2-i \end{pmatrix} \text{ is the eigenvector for } \lambda = 4+i.$$

Thus

$$y = G_1\left[\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cos t - \begin{pmatrix} 0 \\ -1 \end{pmatrix} \sin t\right] e^{4t}$$

$$+ G_2\left[\begin{pmatrix} 0 \\ -1 \end{pmatrix} \cos t + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \sin t\right] e^{4t}$$

$$= G_1 \begin{pmatrix} \cos t \\ 2\cos t + \sin t \end{pmatrix} e^{4t} + G_2 \begin{pmatrix} \sin t \\ -\cos t + 2\sin t \end{pmatrix} e^{4t}$$

(2). $$\begin{vmatrix} -\lambda & 0 & 1 \\ 0 & -\lambda & -1 \\ 0 & 1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda[\lambda^2 + 1] = 0$$

$$\Rightarrow \lambda = 0, \quad \lambda = \pm i$$

For $\lambda = 0$, we have

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right)$$

$\Rightarrow u_2 = 0$ and $u_3 = 0$

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} p \\ 0 \\ 0 \end{pmatrix}, \quad p \in \mathbb{R}$$

$\therefore \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ is the eigenvector for $\lambda = 0$

For $\lambda = i$, we have

$$\left(\begin{array}{ccc|c} -i & 0 & 1 & 0 \\ 0 & -i & -1 & 0 \\ 0 & 1 & -i & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1/i & 0 \\ 0 & 1 & 1/i & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$\Rightarrow v_1 - \frac{1}{i} v_3 = 0 \quad \Rightarrow i v_1 = v_3 \quad \Rightarrow v_1 = -i v_3$

and $v_2 + \frac{1}{i} v_3 = 0 \quad \Rightarrow -i v_2 = v_3 \quad \Rightarrow v_2 = i v_3$

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} -ip \\ ip \\ p \end{pmatrix}, \quad p = 0$$

$\therefore \begin{pmatrix} -i \\ i \\ 1 \end{pmatrix}$ is the eigenvalue for $\lambda = i$

$$\therefore y = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \left[\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cos t - \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \sin t \right] + c_3 \left[\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \cos t + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \sin t \right]$$

$$= c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} \sin t \\ -\sin t \\ \cos t \end{pmatrix} + c_3 \begin{pmatrix} -\cos t \\ \cos t \\ \sin t \end{pmatrix}$$

Nonhomogeneous Linear Systems

$$y' = Ay + f(t) \quad , f(t) \neq 0$$

Undetermined Coefficients

Recall that this method is only applicable if the entries A are constants, and entries of f(t) are constants, polynomials, exponential functions, sines and cosines, or finite sums and products of these functions.

Examples

Find a particular solution for the system

① $y' = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} y + \begin{bmatrix} -7 \\ 5 \end{bmatrix}$

② $y' = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{bmatrix} y + \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} e^{4t}$

Soln

① $y_p = \begin{pmatrix} A \\ B \end{pmatrix} \Rightarrow y_p' = 0$

Substituting into the system, we get

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} + \begin{bmatrix} -7 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2A + 3B - 7 \\ -A - 2B + 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\Rightarrow 2A + 3B - 7 = 0$ and $A = -2B + 5$

$\Rightarrow -4B + 10 + 3B - 7 = 0$

$\Rightarrow B = 3$, $\Rightarrow A = -1$

$\therefore y_p = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

② $y_p = \begin{bmatrix} A \\ B \\ C \end{bmatrix} e^{4t} \Rightarrow y_p' = \begin{bmatrix} A \\ B \\ C \end{bmatrix} 4e^{4t}$

Substitution into the system gives

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} Ae^{4t} \\ Be^{4t} \\ Ce^{4t} \end{bmatrix} + \begin{bmatrix} e^{4t} \\ -e^{4t} \\ 2e^{4t} \end{bmatrix} = \begin{bmatrix} 4Ae^{4t} \\ 4Be^{4t} \\ 4Ce^{4t} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} Ae^{4t} + Be^{4t} + Ce^{4t} + e^{4t} \\ 2Be^{4t} + 3Ce^{4t} - e^{4t} \\ 5Ce^{4t} + 2e^{4t} \end{bmatrix} = \begin{bmatrix} 4Ae^{4t} \\ 4Be^{4t} \\ 4Ce^{4t} \end{bmatrix}$$

$\Rightarrow A + B + C + 1 = 4A$

$2B + 3C - 1 = 4B$

$5C + 2 = 4C \quad \Rightarrow C = -2$, then

$2B - 7 = 4B \Rightarrow B = -7/2$

and $A - 7/2 - 2 + 1 = 4A \Rightarrow A = -3/2$

$\therefore y_p = \begin{pmatrix} -3/2 \\ -7/2 \\ -2 \end{pmatrix} e^{4t}$

VARIATION OF PARAMETERS

FUNDAMENTAL MATRIX

Let $y_1, y_2, \ldots, y_n$ be a fundamental set of solutions of $y' = Ay$ on interval I. The general solution is the linear combination $y = C_1 y_1 + C_2 y_2 + \ldots + C_n y_n$ or

$$y = C_1 \begin{pmatrix} y_{11} \\ y_{21} \\ \vdots \\ y_{n1} \end{pmatrix} + C_2 \begin{pmatrix} y_{12} \\ y_{22} \\ \vdots \\ y_{n2} \end{pmatrix} + \ldots + C_n \begin{pmatrix} y_{1n} \\ y_{2n} \\ \vdots \\ y_{nn} \end{pmatrix} = \begin{pmatrix} C_1 y_{11} + C_2 y_{12} + \ldots + C_n y_{1n} \\ C_1 y_{21} + C_2 y_{22} + \ldots + C_n y_{2n} \\ \vdots \\ C_1 y_{n1} + C_2 y_{n2} + \ldots + C_n y_{nn} \end{pmatrix}$$

$$\Rightarrow y = \begin{pmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nn} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

$$= \Phi(t)\, C$$

The matrix $\Phi(t)$ is a fundamental matrix of $y' = Ay$ on I.

Properties of fundamental matrix $\Phi(t)$

① A fundamental matrix is nonsingular, i.e its determinant is nonzero

② A fundamental matrix of $y' = Ay$ satisfies

$$\Phi'(t) = A\,\Phi(t)$$

A particular solution of the nonhomogeneous system $y' = Ay + f(t)$ is obtained by letting $y_p = \Phi(t)\,u(t)$, where $u(t) = \int \Phi^{-1}(t) f(t)\, dt$. i.e $y_p = \Phi(t) \int \Phi^{-1}(t) f(t)\, dt$.

Examples

Solve

① $y' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} y + \begin{pmatrix} \sec t \\ 0 \end{pmatrix}$

② $y' = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} y + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} e^t$

Sol'n ① $\begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 = 0 \quad \Rightarrow \lambda = \pm i$

For $\lambda = i$, we have

$$\left(\begin{array}{cc|c} -i & -1 & 0 \\ 1 & -i & 0 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & i & 0 \\ 0 & 0 & 0 \end{array}\right) \Rightarrow u_1 + iu_2 = 0$$

$$\Rightarrow -ui = u_2$$

$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} p \\ -ip \end{pmatrix}$, $p \in \mathbb{R}$

Taking $p=1$, eigenvector is $\begin{pmatrix} 1 \\ -i \end{pmatrix}$

$$y_c = c_1\left[\begin{pmatrix}1\\0\end{pmatrix}\cos t - \begin{pmatrix}0\\-1\end{pmatrix}\sin t\right] + c_2\left[\begin{pmatrix}0\\-1\end{pmatrix}\cos t + \begin{pmatrix}1\\0\end{pmatrix}\sin t\right]$$

$$= c_1\begin{pmatrix}\cos t\\ \sin t\end{pmatrix} + c_2\begin{pmatrix}\sin t\\ -\cos t\end{pmatrix}$$

$$\Phi(t) = \begin{pmatrix}\cos t & \sin t\\ \sin t & -\cos t\end{pmatrix}, \quad \Phi^{-1}(t) = \begin{pmatrix}\cos t & \sin t\\ \sin t & -\cos t\end{pmatrix}$$

$$y_p = \Phi(t)\int \Phi(t) f(t)\,dt$$

$$= \begin{pmatrix}\cos t & \sin t\\ \sin t & -\cos t\end{pmatrix}\int\begin{pmatrix}\cos t & \sin t\\ \sin t & -\cos t\end{pmatrix}\begin{pmatrix}\sec t\\ 0\end{pmatrix}dt$$

$$= \begin{pmatrix}\cos t & \sin t\\ \sin t & -\cos t\end{pmatrix}\int\begin{pmatrix}1\\ \tan t\end{pmatrix}dt$$

$$= \begin{pmatrix}\cos t & \sin t\\ \sin t & -\cos t\end{pmatrix}\begin{pmatrix}t\\ -\ln|\cos t|\end{pmatrix}$$

$$= \begin{pmatrix}t\cos t - \sin t\ln|\cos t|\\ t\sin t + \cos t\ln|\cos t|\end{pmatrix}$$

$$y = y_c + y_p$$

$$= c_1\begin{pmatrix}\cos t\\ \sin t\end{pmatrix} + c_2\begin{pmatrix}\sin t\\ -\cos t\end{pmatrix} + \begin{pmatrix}t\cos t - \sin t\ln|\cos t|\\ t\sin t + \cos t\ln|\cos t|\end{pmatrix}$$

$$= c_1\begin{pmatrix}\cos t\\ \sin t\end{pmatrix} + c_2\begin{pmatrix}\sin t\\ -\cos t\end{pmatrix} + \begin{pmatrix}\cos t\\ \sin t\end{pmatrix}t + \begin{pmatrix}-\sin t\\ \cos t\end{pmatrix}\ln|\cos t|$$

② $\begin{vmatrix}1-\lambda & -1\\ 1 & 1-\lambda\end{vmatrix} = (1-\lambda)^2 + 1 = 0$

$\Rightarrow \lambda^2 - 2\lambda + 2 = 0$

$\Rightarrow \lambda = 1 \pm i$

For $\lambda = 1+i$, we have

$$\left(\begin{array}{cc|c} -i & -1 & 0 \\ 1 & -i & 0 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & \frac{1}{i} & 0 \\ 0 & 0 & 0 \end{array}\right) \Rightarrow u_1 + \tfrac{1}{i}u_2 = 0$$
$$\Rightarrow -iu_1 = u_2$$

thus eigenvector is $\begin{pmatrix} 1 \\ -i \end{pmatrix}$

$$y_c = c_1\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}\cos t - \begin{pmatrix} 0 \\ -1 \end{pmatrix}\sin t\right]e^t + c_2\left[\begin{pmatrix} 0 \\ -1 \end{pmatrix}\cos t + \begin{pmatrix} 1 \\ 0 \end{pmatrix}\sin t\right]e^t$$
$$= c_1\begin{pmatrix} \cos t \\ \sin t \end{pmatrix}e^t + c_2\begin{pmatrix} \sin t \\ -\cos t \end{pmatrix}e^t$$

$$\Phi(t) = \begin{pmatrix} (\cos t)e^t & e^t\sin t \\ e^t\sin t & -e^t\cos t \end{pmatrix}, \quad \Phi^{-1}(t) = \frac{-1}{e^{2t}}\begin{pmatrix} -e^t\cos t & -e^t\sin t \\ -e^t\sin t & e^t\cos t \end{pmatrix}$$
$$= \frac{1}{e^t}\begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}$$

$$y_p = \Phi(t)\int \Phi^{-1}(t)\, f(t)\, dt$$
$$= \begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}e^t \int \frac{1}{e^t}\begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}\begin{pmatrix} \cos t \\ \sin t \end{pmatrix}e^t\, dt$$
$$= \begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}e^t \int \begin{pmatrix} 1 \\ 0 \end{pmatrix} dt$$
$$= e^t\begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix}\begin{pmatrix} t \\ 0 \end{pmatrix}$$
$$= e^t\begin{pmatrix} t\cos t \\ t\sin t \end{pmatrix}$$

$$\therefore y = y_c + y_p$$
$$= c_1\begin{pmatrix} \cos t \\ \sin t \end{pmatrix}e^t + c_2\begin{pmatrix} \sin t \\ -\cos t \end{pmatrix}e^t + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}te^t$$